S8. Recomputation record of the c1 findings

This supplement gives, for every finding that was classed c1 at some stage, the value printed in the document, the value computed by the program and the value of the recomputation by a person who had not seen the code (Table S8.1), and, for every settled c1 finding, the inputs from the document, the expression used and the result (Section 3). The complete derivations, with the history of each finding, are in the research log of the repository (`docs/research/research-log.md`; findings F…, decisions D106–D112).

1. How the recomputation was made

The recomputer is a metrologist of the same institute as the author, not an author of the paper, who had not seen the program's code or its results. For each finding, a form gave the document, the example, the printed quantity and the point at which to start (Table S8.2); the start was always the data of the example, not a total or table printed after them, because a recomputation that starts from a printed total reproduces it by construction. The program's values were in a separate key, opened after the form had been filled in. The recomputer worked from the documents alone, with hand calculation, Python (NumPy, SciPy) and Monte Carlo simulation up to 10^7 trials, and wrote every intermediate step on the form. An AI assistant of the Claude family (Anthropic), the family of the assistant with which the program was written, was used in this work; the form does not record which steps it performed. The author checked every written step against the document before comparing the result with the program (research log D108, D109).

The recomputation is therefore independent of the program's code and of its author, but not of the assistant. For this reason Section 3 gives each settled c1 finding as a few lines of arithmetic with the numbers printed in the document, which a reader can repeat by hand.

2. Printed, program and recomputed values

Table S8.1. The 13 findings of the recomputation form (classed c1 when the form was written) and F101, which was split from F40 afterwards. Printed: the value in the document; program: KAPSAM; recomputer: the value written on the form. Class: after the recomputation and the second internal review.

F	Document	Quantity	Printed	Program	Recomputer	Class	Remark
F21	EA-4/02 S13	d_X / mm	90.0003	90.000 23	90.000 23	c1	
F28	NIST TN 1900 E4	v / (m/s)	40.64	57.48	57.48 (u 0.35)	c1	
F39	ISO 21748 C.3	U(flour) / %	12.8	12.6	12.6	c1	
F42	ISO 6143:2025 Ex. 3	u(x) / (µmol/mol)	0.013280	0.014246	0.014246	c1	the recomputer's own errors-in-variables fit
F56	UKAS M3003 K.5	U / mg	0.30	0.31 (u_c 0.1525)	0.31 (u_c 0.1528)	c1	
F57	UKAS M3003 K.3	U / dB	0.049	0.040	0.040	c1	
F58	UKAS M3003 K.6	U / nm	81.5	82.04	82.06	c1	82.0 and 82.1 at the printed precision; effect on U 0.0066 and 0.0069 of U

F	Document	Quantity	Printed	Program	Recomputer	Class	Remark
F77	GUM-5 Ex. 15	MCM s.d. / kN	10.1	9.93	9.93 (t, ν = 5); 9.91 (normal)	c1	
F78	GUM-5 Ex. 1	pH_X	7.0109	7.0043	7.0043	c1	
F82	GUM-5 Ex. 4	u(y_CO_2) / (µmol/mol)	5	3.96 → 4	3.91 → 4	c1	program from Table 4.2, recomputer from Table 4.1
F40	ISO 5725-3:2023 J.5, level 6	s_R / (% × 10^{-3})	15.962	16.331	15.962, 16.331 or 16.781	m	three defensible treatments of a negative variance (Section 4)
F87	EMUE E3.4	U(Pb) / (mg/kg)	2.8	2.9	2.87 → 2.9	c2	with u′ rounded to 5 %, as the example rounds, 2.8 follows too
F90	EMUE E3.3	u (validation) / (nmol/mol)	4.7	4.52	4.47	open	the printed (rounded) inputs allow 4.14–4.78
F101	ISO 5725-3:2023 J.2/J.5, level 5	s_r / (% × 10^{-3})	6.237	14.715 (2023 data); 6.237 (1994 data)	not reproducible from the 2023 data	c1	the cause is the data table

The recomputer's value equals the program's at the printed digits in all rows except F40, where it gives the three treatments, F82, where the two start from different tables of the document and agree after rounding, and F90, where both lie inside the range that the printed inputs allow.

Table S8.2. Where the form asked the recomputation to start.

F	Start
F21	Δl as the mean of the five observations and the coaxiality correction from Eq. (S13.5); not the values entered in the budget S13.10.
F28	The equation printed in E4 with the printed Δ, p, T and R_s, evaluated by hand; not the NUM example file.
F39	The flour components of Table C.4, combined by the formula of C.3.
F40	The raw data of Table J.2 with the stated outlier exclusions; not the summary in Table J.5.
F42	Formula (15) for y = 4950.6 with the parameters and their covariances of the example; not Table B.12.
F56	Each u_i of K.5.9 from its stated limit and distribution, then their combination; not the printed u_c.
F57	The mismatch term from formula E5 (E.9.5) with the coefficients of K.3.6, then u_c and U.
F58	The sensitivity coefficients from the model of K.6.1, −L(α·Δt + Δα·ΔT), then u_c and U.
F77	An own Monte Carlo evaluation with the input distributions stated in the example.
F78	The measured data of the example and its measurement model; not the pH value printed in the result.
F82	The input quantities of the example (Table 4.1), propagated by the law of propagation of uncertainty.
F87	The column u(δ̂)/x_ref of Table E3.4.2 summarised as the text says (root mean square), then U for Pb.
F90	Eq. (E3.3.19) with the inputs of the example; the result to three significant digits.

3. Derivations of the settled c1 findings

The criterion (Section 4.3 of the paper): the inconsistency, once corrected, changes the final printed result at its printed digits, even with the document's own rounding.

F21. EA-4/02 S13 (gauge block, diameter of a ring). Data: five observations of Δl with mean 49.999 536 mm; the coaxiality correction of Eq. (S13.5), δl_P = −0.0037 µm. The budget S13.10 enters Δl = 49.999 55 mm and δl_P = +0.000 004 mm and obtains d_X = 40.0007 + 49.999 55 + 0.000 004 = 90.000 25 mm, reported as (90.000 3 ± 0.000 9) mm. With the data: d_X = 40.0007 + 49.999 536 − 0.000 004 = 90.000 232 mm → (90.000 2 ± 0.000 9) mm. The printed digit changes (0.1 µm = 0.11 U).

F28. NIST TN 1900 E4 (Pitot tube). Stated equation v = √(2ΔR_sT/p) with Δ = 1.993 kPa, p = 101.4 kPa, T = 292.8 K, R_s = 287.058 J kg^{-1} K^{-1}: v = √(2 × 1993 × 287.058 × 292.8 / 101 400) = 57.48 m/s. Printed v = 40.64 m/s = 57.48/√2, the value of √(ΔR_sT/p); the printed u(v) = 0.25 m/s and the intervals scale in the same way (relative uncertainty 0.61 % in both). Reported independently by Delker (2021, footnote 1).

F39. ISO 21748 C.3 (flour). Table C.4 components 2.4 %, 5.0 % and 3.0 %: u = √(2.4² + 5.0² + 3.0²) = 6.31 % (6.29 % with the unrounded s_L), U = 2u = 12.6 %. Printed u = 6.4 %, U = 12.8 %; the intervals of Table C.5 follow from 12.8 % (with 12.6 % the flour interval is 80 to 282 CFU, not 79 to 285). Shrimp (7.8 %) and vegetables (8.9 %) are reproduced from the same table.

F42. ISO 6143:2025 Example 3 (exponential function). The standard states (B.1, B.2.4) that its program B_LEAST gives incorrect parameter uncertainties for the exponential function and prints them (u(b) = 8.1404 × 10^{-2}, 7.3965 × 10^{-2}, 3.7068 × 10^{-11}); the errors-in-variables fit of the printed calibration data gives 3.18, 3.17 and 1.274 × 10^{-6} (identical with ODRPACK). Formula (15) for the sample at y = 4950.6 with the correct covariance matrix gives u(x) = 0.014 246 µmol/mol; Table B.12 prints 0.013 280, which follows from the printed (incorrect) covariances (0.013 283). The printed u(x) is 6.8 % too low; the standard does not correct it. A Monte Carlo simulation of the calibration (4000 refits) gives 0.014 06.

F56. UKAS M3003 K.5 (mass). The u_i column of K.5.9: 0.05, 0.058, 0.029, 0.029, 0.115, 0.05 mg. Σu_i² = 0.0025 + 0.003 364 + 0.000 841 + 0.000 841 + 0.013 225 + 0.0025 = 0.023 271 mg², u_c = 0.1525 mg, U = 2u_c = 0.305 → 0.31 mg. Printed u_c = 0.150 mg, U = 0.30 mg. The printed u_c follows if the drift limit is divided by 2 instead of √3, or if one of the two rounding terms is left out; the cause cannot be identified.

F57. UKAS M3003 K.3 (attenuator, mismatch). Formula E5 of E.9.5 with the coefficients of K.3.6 (Γ_G = Γ_L = 0.03, s11a = s11b = 0.05, s22a = 0.05, s22b = 0.01, s21a = 1, s21b = 0.31) gives a mismatch limit M = 0.0170 dB; the budget uses 0.022 dB. With 0.0170 dB and the other components of the budget unchanged, u_c = 0.0202 dB and U = 0.040 dB; printed U = 0.049 dB. (0.022 dB follows with s11 = 0.09 for one of the coefficients; not identifiable from the example.)

F58. UKAS M3003 K.6 (gauge block). The model contains −L(α·Δt + Δα·ΔT). By the model, the temperature difference Δt = ±0.08 °C acts through L·α = 115 nm/°C: ±9.2 nm, rectangular, u = 5.3 nm; the product Δα·ΔT has zero first-order sensitivity and contributes L·u(Δα)·u(ΔT) = 0.94 nm through the second-order term. The budget instead evaluates Δt through Δα (±1.6 nm) and adds it linearly to the ΔT term (±4 nm) as "correlated" (±5.6 nm, triangular). With the model's coefficients, u_c = 41.0 nm and U = 82.0 nm; printed u_c = 40.7 nm, U = 81.5 nm (0.6 % of U). EA-4/02 S4 evaluates the same model as the stated model requires.

F77. JCGM GUM-5 Example 15 (wire rope). 15.3.1 assigns to F a t-distribution with mean 10 005.7 kN, scale $u(\bar{F}) = s/\sqrt{6} = 0.7$ kN and $\nu = 5$. The Monte Carlo standard deviation is then $\sqrt{u_c^2 - u^2(\bar{F}) + u^2(\bar{F})\cdot\nu/(\nu - 2)} = 9.93$ kN (10^6 trials: 9.93 kN); 15.5.2 prints 10.1 kN. 10.1 kN follows if the scale is the standard deviation of one test, s = 1.75 kN, instead of that of the mean. The difference 9.9 against 10.1 kN that 15.6.1 attributes to the shape of the distribution is this input inconsistency.

F78. JCGM GUM-5 Example 1 (pH). 1.9.1 obtains $pH_{X,T1} = 7.0002$ and $pH_{X,T2} = 7.0161$ and, by the third interpolation, $pH_X = 7.0043$; it then states "$pH_X = 7.0109$, $u(pH_X) = 0.0041$". 7.0109 does not follow from the data; the document prints the correct intermediate value 7.0043 itself (0.80 U).

F82. JCGM GUM-5 Example 4 (gas mixture). The law of propagation of uncertainty with the inputs of Table 4.2 (20 inputs; dominated by the atomic weights of N, C and O) gives $u(y_{CO_2}) = 3.96$ µmol/mol, which rounds to 4; with the means and standard deviations of the means of Table 4.1 it gives 3.91 → 4. Printed: 5 µmol/mol (u_{rel} 0.003 %, recomputed 0.0020 %). With $u(\hat{r}_j) = 2$ mg for the three weighings the result would be 4.50 → 5; the document does not state such a value. $u(y_{N_2}) = 4.72$ → 5 agrees with the printed value.

F101. ISO 5725-3:2023 Tables J.2 and J.5 (vanadium in steel). Table J.2 of the 2023 edition differs from Table D.2 of the 1994 edition in 15 entries: 12 at level 3 (the second day-1 result of laboratories 7–20 repeats the first), 1 at level 4 (laboratory 18) and 2 at level 5 (laboratories 2 and 3). Table J.5, carried over unchanged, is reproduced exactly from the 1994 data with the stated exclusions, but not from the 2023 Table J.2: at level 5, $s_r = 14.715 \times 10^{-3}$ % from the 2023 data against the printed 6.237×10^{-3} %; at level 3, 0.652 against 1.739. The printed results do not follow from the document's own data; the cause is the data table.

4. Findings reclassified or open

F40 (class m). At level 6 of ISO 5725-3 J.5, the mean squares $MS_0 = 617.90$, $MS_1 = 55.39$ and $MS_e = 91.11$ ($\times 10^{-6}$) give a negative day variance $s_{(1)}^2 = -26.79 \times 10^{-6}$. The printed $s_R = 15.962 \times 10^{-3}$ % $= \sqrt{((MS_0 + MS_1 + MS_e)/3)}$ applies the formulas as printed and equals the REML estimate with variances ≥ 0; 16.331 sets $s_{(1)}^2 = 0$ throughout (the program); 16.781 sets it to zero in the sum only (ISO 5725-2, 8.4.5.4). All three are defensible, so the finding concerns the consistency of the printed row ($s_I(T) = s_r$ with $s_{(1)}^2 = 0$, s_R with it kept), not a final result.

F87 (class c2). The text calls 3.3 % the root mean square of the relative bias uncertainties of Table E3.4.2; it is their arithmetic mean (3.28 %), the root mean square is 3.50 %. With x = 27.8 mg/kg, U = 2.87 → 2.9 mg/kg; with u′ rounded to 5 %, as the example rounds it, the printed 2.8 mg/kg follows either way.

F90 (open). Eq. (E3.3.19) with the printed (rounded) inputs gives values between 4.14 and 4.78 nmol/mol, because three large terms cancel; the printed 4.7 lies in this range. With the unrounded parameters of a full weighted total least-squares fit, u = 4.52 (4.35 without the covariances). Whether the printed 4.7 follows from the document's own unrounded values cannot be decided from the example; the finding is open and not counted.

5. Errata and amendments of the documents with class-c findings

Checked on 2026-11-11 (research log D112), before submission, for every document with a settled c1 finding and for the other documents of class-c findings that the article discusses. "None found" means: no erratum, corrigendum, amendment or later edition listed by the issuing body on the date of the check; it

does not mean that none exists in a form the issuing body does not list (a national adoption, a committee draft, a private communication).

Document (edition used)	Findings	Where checked	Result
EA-4/02 M:2022 (rev03, 4 April 2022)	F21 (c1)	EA publications page (Documents, EA-4/02)	No erratum. The 2022 file was republished after an editorial error on p. 25 (F4), which does not concern S13; the current file is the one used.
NIST TN 1900 (2015)	F28 (c1), F29 (open)	NIST publication page of TN 1900 (DOI 10.6028/NIST.TN.1900)	No erratum; the page reads “Created November 4, 2015, Updated June 2, 2021” (an update of the page, not of the note). F28 was reported independently by Delker (2021, footnote 1).
ISO 21748:2017	F39 (c1)	ISO page of the standard (status, amendments, corrigenda)	No amendment or corrigendum listed.
ISO 5725-3:2023	F40 (m), F101 (c1)	ISO page of the standard	No amendment or corrigendum. Corrigendum 1 to the 1994 edition (2001) is incorporated in the 2023 edition and concerns neither finding.
ISO 6143:2025	F41 (c2), F42 (c1)	ISO page of the standard; ISO/TC 158 work programme	No corrigendum. ISO 6143:2025/DAmd 1, “Software B_LEAST — Performance check for non-linear model functions using alternative computer programs”, is at the DIS stage (registered 29 September 2026); its text is not public. B.2.4 of the 2025 edition already states that B_LEAST gives incorrect parameter uncertainties for the exponential function (the acknowledged part of F42); whether the amendment also changes the u(x) of Table B.12 can be checked only through a national mirror committee or after publication.
UKAS M3003 edition 6 (March 2024)	F56, F57, F58 (c1)	UKAS publications page of M3003	No erratum or later edition.
JCGM GUM-5:2026	F77, F78, F82 (c1), F79, F80 (m), F81 (c2), F85 (open)	BIPM page of JCGM GUM-5 (DOI 10.59161/YNLY8209)	No erratum listed; the page lists the document and its DOI only.
EMPIR EMUE compendium (Zenodo record 5142180, version 2, Compendium_M36)	F87 (c2), F89 (c2), F90 (open)	Zenodo record (all versions)	Version 2 is the latest version and the one used; no erratum.
IUPAC Technical Report 2024; BIPM ISRD-01	F92, F97 (open)	—	Not checked: open findings, not counted.
Eurachem/CITAC QUAM:2012	F10, F12 (partly erratum), F14 (erratum)	Eurachem page of the guide (QUAM_ERRATA_v2, 21 January 2019)	The errata correct F14 fully and F10, F12 in part (as stated with each finding).

Known before the audit, from a source other than an erratum: F28 (Delker 2021) and the parameter uncertainties of F42 (stated by the standard itself, B.2.4). Not found as known: F21, F39, F56, F57, F58, F77, F78, F82, F101 and the u(x) part of F42.

Not checked: national adoptions (e.g. TS EN ISO), committee drafts other than the ISO 6143 amendment, and private communications to the issuing bodies by other users.